

ToyWorld – E-commerce Database Management System

Chapter 4

INTRODUCTION

The ToyWorld – E-commerce Database Management System is designed to efficiently manage customers, products, categories, shopping carts, orders, payments, shipments, and administrators. Functional dependencies define the relationships between attributes in each table, while redundancy analysis and data anomaly detection help maintain data consistency and reduce unnecessary duplication. A properly normalized database improves performance, ensures data integrity, and supports efficient online shopping operations.

1. CUSTOMER TABLE

Attribute	Key Type
Customer_ID	Primary Key
Name	Not Null
Email	Unique Key
Phone	Unique Key
Address	Not Null

Functional Dependency

Customer_ID → Name, Email, Phone, Address

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders, carts, and reviews store only the Customer_ID, preventing duplicate customer records.

Data Anomalies

- Insert Anomaly: A new customer can be added without placing an order.
- Update Anomaly: Updating a customer's phone number or address requires changes only in the Customer table.
- Delete Anomaly: Deleting an order or review will not remove customer information.

2. CATEGORY TABLE

Attribute	Key Type

Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Functional Dependency

Category_ID → Category_Name, Description

Redundancy Analysis

Each category is stored only once. Products reference the category using Category_ID instead of repeating the category name.

Data Anomalies

- Insert Anomaly: New categories can be added without products.
- Update Anomaly: Renaming a category is done only once.
- Delete Anomaly: Deleting a product does not delete the category.

3. PRODUCT TABLE

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Category_ID	Not Null
Price	Not Null
Stock	-
Brand	-
Description	-

Functional Dependency

Product_ID → Product_Name, Category_ID, Price, Stock, Brand, Description

Redundancy Analysis

Product details are stored only in the Product table. Order items, cart items, and reviews use Product_ID instead of repeating product information.

Data Anomalies

- Insert Anomaly: New products can be added before receiving orders.
- Update Anomaly: Product price or stock is updated only once.

- Delete Anomaly: Deleting an order does not remove product information.

4. ORDER TABLE

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	-
Total_Amount	-

Functional Dependency

Order_ID → Customer_ID, Order_Date, Order_Status, Total_Amount

Redundancy Analysis

The Order table stores only order-related information. Customer details are referenced using Customer_ID.

Data Anomalies

- Insert Anomaly: Orders can be created for registered customers.
- Update Anomaly: Order status can be updated without affecting other tables.
- Delete Anomaly: Deleting an order does not remove customer information.

5. ORDER_ITEM TABLE

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-

Functional Dependency

OrderItem_ID → Order_ID, Product_ID, Quantity

Redundancy Analysis

This table connects orders and products, avoiding repeated product information in the Order table.

Data Anomalies

- Insert Anomaly: Products can be added to an order independently.

- Update Anomaly: Quantity changes affect only the selected order item.
- Delete Anomaly: Removing an order item does not delete the product.

6. CART TABLE

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_Items	-
Total amount	-

Functional Dependency

Cart_ID → Customer_ID, Total_Items, Total_Amount

Redundancy Analysis

Each customer has a separate cart. Customer information is referenced using Customer_ID, reducing redundancy.

Data Anomalies

- Insert Anomaly: A cart can be created before adding products.
- Update Anomaly: Cart details are updated independently.
- Delete Anomaly: Deleting a cart does not remove customer information.

7. PAYMENT TABLE

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Redundancy Analysis

Payment information is stored separately from the Order table to avoid duplicate payment records.

Data Anomalies

- Insert Anomaly: Payment details can be recorded after creating an order.
- Update Anomaly: Payment status is updated only once.
- Delete Anomaly: Deleting payment information does not remove the order.

8. REVIEW TABLE

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	-
Review_Text	-
Review_Date	-

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

Reviews are stored separately and linked to customers and products using foreign keys, avoiding duplicate review information.

Data Anomalies

- Insert Anomaly: A customer can submit a review after purchasing a product.
- Update Anomaly: Review details can be updated independently.
- Delete Anomaly: Deleting a review does not remove customer or product information.

9. SHIPMENT TABLE

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key
Delivery_Address	-
Shipment_Date	-

Delivery_Date	-
Shipment_Status	-

Functional Dependency

Shipment_ID → Order_ID, Delivery_Address, Shipment_Date, Delivery_Date, Shipment_Status

Redundancy Analysis

Shipment details are maintained in a separate table, preventing repeated delivery information.

Data Anomalies

- Insert Anomaly: Shipment records can be added after order confirmation.
- Update Anomaly: Shipment status can be updated independently.
- Delete Anomaly: Deleting shipment information does not affect order details.

Conclusion

The ToyWorld – E-commerce Database Management System is designed using proper functional dependencies and normalized tables. Each entity stores only its relevant data, reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains integrity, and ensures efficient database management for the ToyWorld e-commerce system.